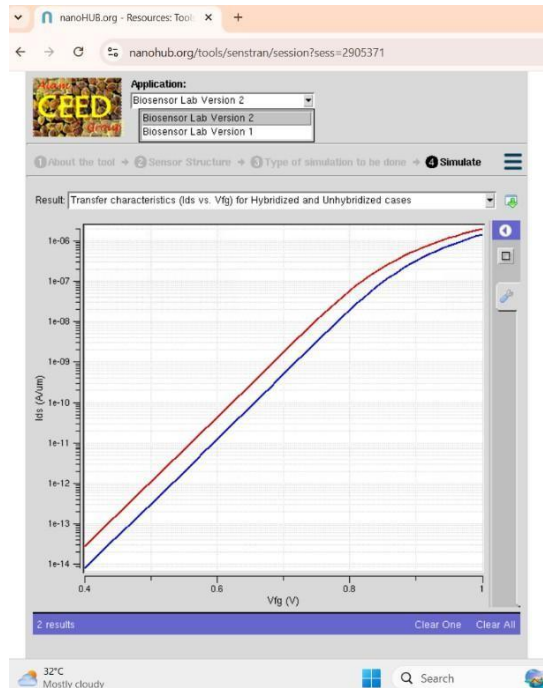


**R. RUPASREE**

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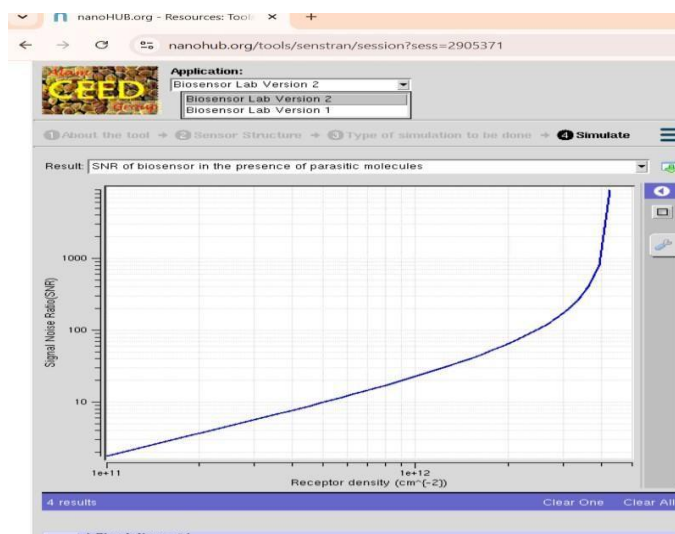
### **Exp. 10. Biosensor and Bio-FET characteristics using NanoHub**

#### **Case Study 1: Comparative Analysis of Hybridized and Unhybridized BioFET Transfer Characteristics for Enhanced Biosensing Performance**



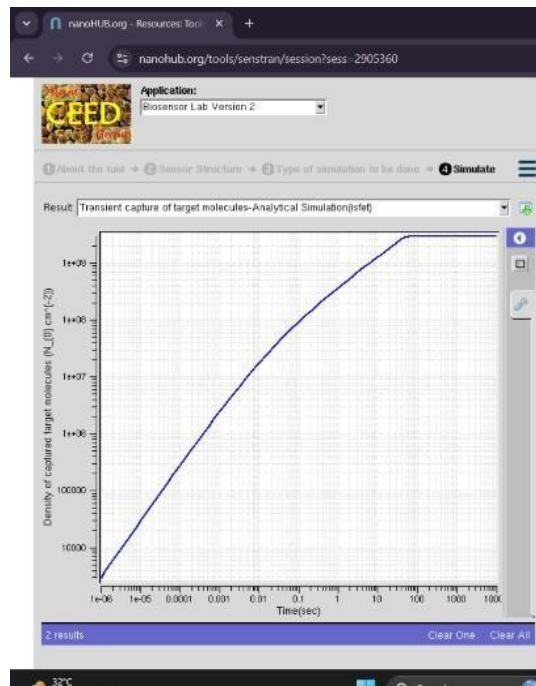
**Outcome:** On an industrial scale the film deposition is done by magnetron sputter deposition; on a laboratory scale molecular beam epitaxy, pulsed laser deposition and electron beam physical vapor deposition are also utilized. The junctions are prepared by photolithography.

#### **Case Study 2: Evaluation of Signal-to-Noise Ratio (SNR) Variation with Receptor Density in the Presence of Parasitic Molecules**



**Outcome:** The simulated VI curves show a relatively high drain current ( $I_{ds}$ ) for a given gate-source voltage ( $V_{gs}$ ), demonstrating efficient carrier transport in the graphene channel. High carrier mobility ( $2000 \text{ cm}^2/\text{Vs}$ ) minimizes scattering effects, enabling rapid carrier movement and improved current conduction.

### Case Study 3: Transient Analysis of Target Molecule Capture Dynamics in a BioFET-Based Biosensor



**Outcome:** The simulated VI curves show a relatively high drain current ( $I_{ds}$ ) for a given gate-source voltage ( $V_{gs}$ ), demonstrating efficient carrier transport in the graphene channel. High carrier mobility ( $2000 \text{ cm}^2/\text{Vs}$ ) minimizes scattering effects, enabling rapid carrier movement and improved current conduction.

